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Maximum likelihood network topology identification from edge-based unicast measurements

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Abstract

Network tomography is a process for inferring "internal" link-level delay and loss performance information based on end-to-end (edge) network measurements. These methods require knowledge of the network topology; therefore a first crucial step in the tomography process is topology identification. This paper considers the problem of discovering network topology solely from host-based, unicast measurements, without internal network cooperation. First, we introduce a novel delay-based measurement scheme that does not require clock synchronization, making it more practical than other previous proposals. In contrast to methods that rely on network cooperation , our methodology has the potential to identify layer two elements (provided they are logical topology branching points and induce some measurable delay). Second, we propose a maximum penalized likelihood criterion for topology identification. This is a global optimality criterion, in contrast to other recent proposals for topology identification that employ suboptimal, pair-merging strategies. We develop a novel Markov Chain Monte Carlo (MCMC) procedure for rapid determination of the most likely topologies. The performance of our new probing scheme and identification algorithm is explored through simulation and Internet experiments.

Real-World Impact & Application

Mapping the Internet from Your Laptop: Early network mapping required special multicast broadcast protocols that home Wi-Fi and commercial ISPs blocked. This paper showed how standard packet pings sent from an ordinary laptop can reconstruct the shape of the internet between servers.